

Part 1 of 4

Defence Lasers and Optronic Systems: Role of Electronics

Advances in the technologies of electronics devices and circuitry over the last two decades have been one of the key factors leading to explosive growth and maturity of laser systems designed for military usage in diverse application scenarios. Electronics that go along with a military laser system are much more than a power supply and involve complex technologies. This first part in a four-part series focusses on the role of electronics and the technologies involved in laser and optoelectronics systems intended for better established military applications

DR ANIL K. MAINI AND NAKUL MAINI

There has been a dramatic increase in the commercial, industrial, medical, defence and research applications of lasers in the last 10 to 15 years. During this period, we have not only seen the use of lasers in applications thought of in the early days of lasers getting matured, but we have also witnessed them making their presence felt very effectively in many new areas.

Laser which was rightly called an invention in search of applications or a solution looking for a problem in 70's is today the most widely used and, more importantly, the most diversely exploited piece of equipment. It is undoubtedly one of the greatest inventions of 20th century along with computers, satellites and integrated circuits.

There are two main factors which have played a major role in enhancing the application potential of lasers. The first of course is the development and subsequent arrival on the commercial scene of a large variety of lasers covering a very large portion of electromagnetic spectrum and producing a wide range of output power levels at an affordable price tag.

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